Chapter 8 Advanced Counting Techniques

Several Advanced Counting Techniques:

- Recurrence Relations
- Generating Functions
- The Principle of Inclusion-Exclusion



Chapter 8 Advanced Counting Techniques

8.1 Applications of Recurrence Relations

8.2 Solving Linear Recurrence Relations

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

8.4 Generating Functions

8.5 Inclusion-Exclusion

8.6 Applications of Inclusion-Exclusion



1. Introduction

Problem:

The number of bacteria in a colony doubles every hour. If a colony begins with five bacteria, how many will be present in *n* hours?

A variety of counting problems can be modeled using recurrence relations.



2. Recurrence Relations

【Definition】 A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, a_2, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integers.

$$a_n = f(a_0, a_1, a_2, \dots, a_{n-1}) \quad n \geq n_0$$

For example,

(1) The Fibonacci sequence $a_n = a_{n-1} + a_{n-2}$.

(2) Pascal's recursion for the binomial coefficient is a *two-variable* recurrence equation:

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$



A solution of a recurrence relation **is a sequence if its terms satisfy the recurrence relation.**

〖Example 1〗 Determine whether the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for n = 2,3,4,…,where $a_n = 3n$ for every nonnegative integer n.

Answer the same question where $a_n = 2^n$ and where $a_n = 5$.

Solution :

$a_n = 3n$ ✓

$a_n = 2^n$ ✗

$a_n = 5$ ✓



Note:

Normally, there are infinitely many sequences which satisfy a recurrence relation.

We distinguish them by the *initial conditions*, the values of a_0, $a_1, a_2, \ldots$ to uniquely identify a sequence.

Given a recurrence relation, how many initial conditions are needed to uniquely identify the sequence?

The *degree* of a recurrence relation

$a_n = a_{n-1} + a_{n-8}$ **---- a recurrence relation of degree 8**



For example,

(1) In the Fibonacci recurrence $a_n = a_{n-1} + a_{n-2}$ we must specify a_0 and a_1 .

(2) In Pascal's identity $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$

we must specify $C(1,0)$ and $C(1,1)$.



3. Modeling with Recurrence Relations

〖Example 2〗 *Rabbits and the Fibonacci numbers*

A young pair of rabbits is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. Find a recurrence relation for the number of pairs of rabbits on the island after *n* months, assuming that no rabbits ever die.



Solution :

f_n : The number of pairs of rabbits after n month

$f_1 = 1$ $\quad\quad$ $f_2 = 1$

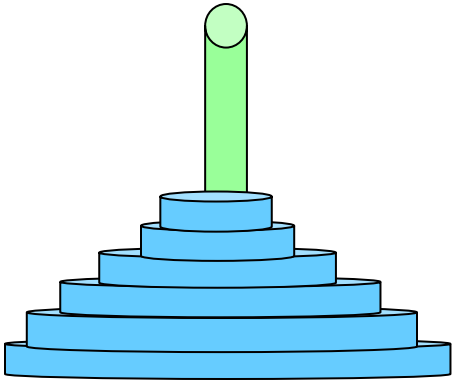
$f_3 = 1+1 = 2$ $\quad\quad$ $f_4 = 2+1 = 3$

……

$f_n = f_{n-1} + f_{n-2}$



〖Example 3〗 *The Tower of Hanoi*



Solution :

Let H_n denote the number o
of Hanoi problem with n d

$H_1, H_2, \ldots, H_n, \ldots$

$$\begin{cases} H_n = 2H_{n-1} + 1 \\ H_1 = 1 \end{cases}$$

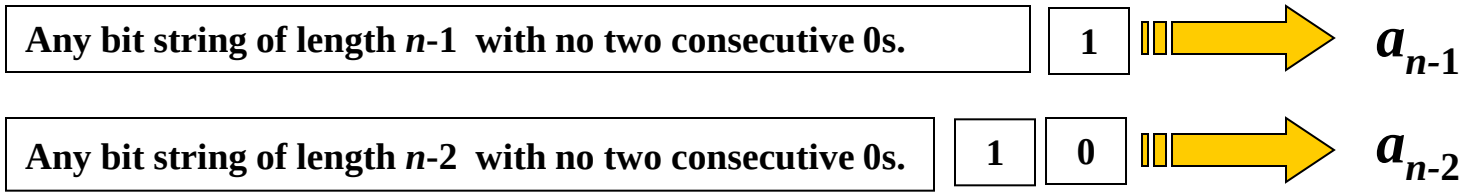
$$\begin{aligned} H_n &= 2H_{n-1} + 1 \\ &= 2(2H_{n-2} + 1) + 1 \\ &= 2^2 H_{n-2} + 2 + 1 \\ &= 2^2(2H_{n-3} + 1) + 2 + 1 \\ &= 2^3 H_{n-3} + 2^2 + 2 + 1 \\ &= \ldots \\ &= 2^{n-1} H_1 + 2^{n-2} + \ldots + 2 + 1 \\ &= 2^{n-1} + 2^{n-2} + \ldots + 2 + 1 \\ &= 2^n - 1 \end{aligned}$$

pproach

〖Example 4〗Find a recurrence relation for the number of bit strings of length *n* that don't have two consecutive 0s.

Solution :

Let a_n denote the number of bit strings of length *n* that don't have two consecutive 0s.



Recurrence relation:

$$a_n = a_{n-1} + a_{n-2}\ ,\ n>=3.$$

Initial conditions: $a_1=2,\ a_2=3$



〖Example 5〗Codeword Enumeration: A computer system considers a string of decimal digits a valid codeword if it contains an even number of 0 digits. Let a_n be the number of valid n-digit codeword. Find a recurrence relation for a_n.

$$a_n = 9\,a_{n-1} + (10^{n-1} - a_{n-1})$$



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The Methods of Solving Recurrence Relations

- **Iterative approach**
- **Use a systematic way to solve an important class of recurrence relations**
- **Generating functions**

Solving recurrence relations can be very difficult unless the recurrence equation has a special form.



Linear homogeneous recurrence relation of degree k with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where c_1, c_2,…, c_k are real numbers, and $c_k \neq 0$

- ***linear***

  - linear combination of previous terms

- ***constant coefficients***

  the coefficients of a_is are constants

- ***degree k***

  a_n is a function of the previous *k* terms of the sequence

- ***homogeneous***

If we put all the a_i s on the left side of the equation and everything else on the right side, then the right side is 0. Otherwise *nonhomogeneous*.



Linear homogeneous recurrence relation of degree k with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where $c_1, c_2, \ldots, c_k$ are real numbers, and $c_k \neq 0$

A sequence satisfying the recurrence relation in the definition is uniquely determined by the recurrence relation and the *k initial conditions*

$$a_0 = C_0,\ a_1 = C_1,\ \ldots,\ a_{k-1} = C_{k-1}$$



〖Example 1〗

(1) $a_n = (1.02)a_{n-1}$

linear; constant coefficients; homogeneous; degree 1

(2) $a_n = (1.02)\, a_{n-1} + 2^{n-1}$

linear; constant coefficients; nonhomogeneous; degree 1

(3) $a_n = a_{n-1} + a_{n-2} + a_{n-3} + 2^{n-3}$

linear; constant coefficients; nonhomogeneous; degree 3

(4) $a_n = n\, a_{n-1} + n^2\, a_{n-2} + a_{n-1}\, a_{n-2}$

nonlinear; coefficients are not constants; homogeneous; degree 2



1. Solving Linear Homogeneous Recurrence Relation With Constant Coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

The basic approach:

To look for solutions of the form $a_n = r^n$, where r is a constant.

Solution Procedure:

(1) Put all a_is on the left side of the equation

$$a_n - c_1 a_{n-1} - c_2 a_{n-2} - \cdots - c_k a_{n-k} = 0$$

(2) Substitute the solution into the equation, factor out the lowest power of r and eliminate it.



$$r^n - c_1 r^{n-1} - c_2 r^{n-2} - \cdots - c_k r^{n-k} = 0$$

$$r^{n-k}(r^k - c_1 r k^{n-1} - c_2 r^{k-2} - \cdots - c_$$

Characteristic equation

(3) We obtain the equivalent equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k = 0$$

Characteristic root

(4) Find its *k* roots $r_1, r_2, \cdots, r_k$

These characteristic roots can be used to give an explicit formula for all the solutions of the recurrence relation.



【Theorem 1】 Let c_1, c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1, r_2. Then the Sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0,1,2,\cdots$, where α_1, α_2 are constants.

Proof:

- **Show that if r_1, r_2 are the roots of the characteristic equation, and α_1, α_2 are constant, then the sequence $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ is a solution of the recurrence relation**

$$r_1^2 = c_1 r_1 + c_2$$

$$r_2^2 = c_1 r_2 + c_2$$



$$c_1 a_{n-1} + c_2 a_{n-2} = c_1(\alpha_1 r_1^{n-1} + \alpha_2 r_2^{n-1}) + c_2(\alpha_1 r_1^{n-2} + \alpha_2 r_2^{n-2})$$

$$= \alpha_1 r_1^{n-2}(c_1 r_1 + c_2) + \alpha_2 r_2^{n-2}(c_1 r_2 + c_2)$$

$$= \alpha_1 r_1^{n} + \alpha_2 r_2^{n}$$

$$= a_n$$

- **Show that if $\{a_n\}$ is a solution, then $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for some constant α_1, α_2.**

  Suppose that $\{a_n\}$ is a solution, and the initial condition $a_0 = C_0, a_1 = C_1$ hold.

$$a_0 = C_0 = \alpha_1 + \alpha_2$$
$$a_1 = C_1 = \alpha_1 r_1 + \alpha_2 r_2$$

$$\Rightarrow \quad \alpha_1 = \frac{C_1 - C_0 r_2}{r_1 - r_2}, \qquad \alpha_2 = \frac{C_0 r_1 - C_1}{r_1 - r_2}$$



〖Example 2〗Find an explicit formula for the fibonacci numbers.

Solution:

$f_n = f_{n-1} + f_{n-2}, \quad f_0 = 0, \quad f_1 = 1$

(1) Determine the characteristic equation: $r^2 - r - 1 = 0$

(2) Find its roots: $r_1 = \frac{1+\sqrt{5}}{2}, r_2 = \frac{1-\sqrt{5}}{2}$

(3) Obtain the general solution:

$$f_n = \alpha_1 (\frac{1+\sqrt{5}}{2})^n + \alpha_2 (\frac{1-\sqrt{5}}{2})^n, \alpha_1, \alpha_2 \text{ are constant.}$$

(4) Dtermine α_1, α_2**:** $\alpha_1 = \frac{1}{\sqrt{5}}, \alpha_2 = -\frac{1}{\sqrt{5}}$

Consequently, the fibonacci numbers are given by

$$f_n = \frac{1}{\sqrt{5}} (\frac{1+\sqrt{5}}{2})^n - \frac{1}{\sqrt{5}} (\frac{1-\sqrt{5}}{2})^n$$



【 Theorem 2】 Let c_1, c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0 . A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$ for $n = 0,1,2,\cdots$, where α_1, α_2 are constants.

Proof:

Omitted.



〖Example 3〗 $a_n = 6a_{n-1} - 9a_{n-2}, a_0 = a_1 = 1$

Solution:

(1) Determine the characteristic equation: $(b-3)^2 = 0$

(2) Find its roots : $b_1 = b_2 = 3$

(3) Compute the general solution: $a_n = (\alpha_1 + \alpha_2 n)3^n$

(4) Solve for coefficients :

$$\begin{cases} a_0 = 1 = \alpha_1 \\ a_1 = 1 = (\alpha_1 + \alpha_2) \cdot 3 \end{cases}$$

Consequently, $a_n = (1 - \frac{2}{3}n)3^n$



【 Theorem 3】 Let $c_1, c_2, \cdots, c_k$ be real numbers. Suppose that the characteristic equation $r^k - c_1 r^{k-1} - \ldots - c_k = 0$ has *k* distinct roots $r_1, r_2, \cdots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \cdots + \alpha_k r_k^n$ for $n = 0,1,2,\cdots$ where $\alpha_1, \alpha_2, \cdots, \alpha_k$ are constants.

The coefficients $\alpha_1, \alpha_2, \cdots, \alpha_k$ are found by enforcing the initial conditions



〖Example 4〗 What is the solution of the recurrence relation

$$a_{n+2} = 3a_{n+1}, a_0 = 4$$

Solution:

(1) The Characteristic equation of the recurrence relation is $r - 3 = 0$.

(2) Find the root of the characteristic equation: $r_1 = 3$

(3) Compute the general solution: $a_n = c3^n$

(4) Find the constants based on the initial conditions:

$$a_0 = c3^n = 4$$

(5) Produce the specific solution: $a_n = 4 \cdot 3^n$



【 Theorem 4】 Let $c_1, c_2, \cdots, c_k$ **be real numbers. Suppose that the characteristic equation** $r^k - c_1 r^{k-1} - \ldots - c_k = 0$ **has *t* distinct roots** $r_1, r_2, \cdots, r_t$ **with multiplicities** $m_1, m_2, \cdots, m_t$ **, respectively, so that** $m_i \geq 1 \; for \quad i = 1,2,\cdots,t$ **and** $m_1 + m_2 + \cdots + m_t = k$ **Then a sequence** $\{a_n\}$ **is a solution of the recurrence relation** $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ **if and only if**

$$a_n = (\alpha_{1,0} + \alpha_{1,1} n + \cdots + \alpha_{1,m_1-1} n^{m_1-1}) r_1^n +$$

$$(\alpha_{2,0} + \alpha_{2,1} n + \cdots + \alpha_{2,m_2-1} n^{m_2-1}) r_2^n + \cdots +$$

$$(\alpha_{t,0} + \alpha_{t,1} n + \cdots + \alpha_{t,m_t-1} n^{m_t-1}) r_t^n$$

for $n = 0,1,2,\cdots$ **where** $\alpha_{i,j}$ **are constants for** $1 \leq i \leq t, 0 \leq j \leq m_i - 1$.



2. Linear Nonhomogeneous Recurrence Relation With Constant Coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$

Where $c_i (i=1,2,\ldots,k)$ is real numbers, $F(n)$ is a function not identically zero depending only on n.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

the associated *homogeneous* recurrence relation

Note:

Solutions to nonhomogeneous case is the sum of solutions to associated homogeneous recurrence system and a particular solution to the nonhomogeneous case.



【 Theorem 5】 Let $\{a_n^{(p)}\}$ be a *particular solution* of the nonhomogeneous linear recurrence relation with constant coefficients $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$
Then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$

Proof:

$$a_n^{(p)} = c_1 a_{n-1}^{(p)} + c_2 a_{n-2}^{(p)} + \cdots + c_k a_{n-k}^{(p)} + F(n)$$

Suppose that $\{b_n\}$ is a second solution of the nonhomogeneous recurrence relation, so that

$$b_n = c_1 b_{n-1} + c_2 b_{n-2} + \cdots + c_k b_{n-k} + F(n)$$

$$b_n - a_n^{(p)} = c_1(b_{n-1} - a_{n-1}^{(p)}) + c_2(b_{n-2} - a_{n-2}^{(p)}) + \cdots + c_k(b_{n-k} - a_{n-k}^{(p)})$$



【 Theorem 6】 Assume a linear nonhomogeneous recurrence equation with constant coefficients with the nonlinear part *F(n)* of the form

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n + b_0) s^n$$

If *s* is not a root of the characteristic equation of the associated homogeneous recurrence equation, there is a particular solution of the form

$$(p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n$$

If *s* is a root of multiplicity *m* , a particular solutions is of the form

$$n^m (p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n$$



〖Example 5〗 $a_n = 6a_{n-1} - 9a_{n-2} + F(n),$

$where F(n) = 3^n, n3^n, n^2 2^n, and (n^2+1)\cdot 3^n$

Solution:

(1) The general solution of the associated homogeneous recurrence equation :

$$a_n = (\alpha_1 + \alpha_2 n)\cdot 3^n$$

(2) A particular solution of the form:

$$p_0 n^2 3^n \ if \ F(n) = 3^n$$

$$n^2(p_1 n + p_0)\cdot 3^n \ if \ F(n) = n3^n$$

$$(p_2 n^2 + p_1 n + p_0)\cdot 2^n \ if \ F(n) = n^2 2^n$$

$$n^2(p_2 n^2 + p_1 n + p_0)\cdot 3^n \ if \ F(n) = (n^2+1)3^n$$



〖Example 6〗Let a_n be the sum of the first n positive integers. Note that a_n satisfies the recurrence relation $a_n = a_{n-1} + n$. Find The explicit formula of a_n

Solution:

(1) The general solution of the associated homogeneous recurrence equation : $a_n^{(h)} = c \cdot (1)^n = c$

(2) A particular solution of the form:

$$n(p_1 n + p_0) = p_1 n^2 + p_0 n$$

(3) Find p_1,p_0:

$$p_1 n^2 + p_0 n = p_1 (n-1)^2 + p_0 (n-1) + n$$

$$n(2p_1 - 1) + (p_0 - p_1) = 0$$

$$p_0 = p_1 = 1/2,$$

(4) Find c: using initial condition a_1=1



Homework: (Due on April 25)

Ver. 8 & Ver. 7

Sec. 8.1: 8, 10, 26, 29, 32

Sec. 8.2 2, 4(g), 20, 30, 35

